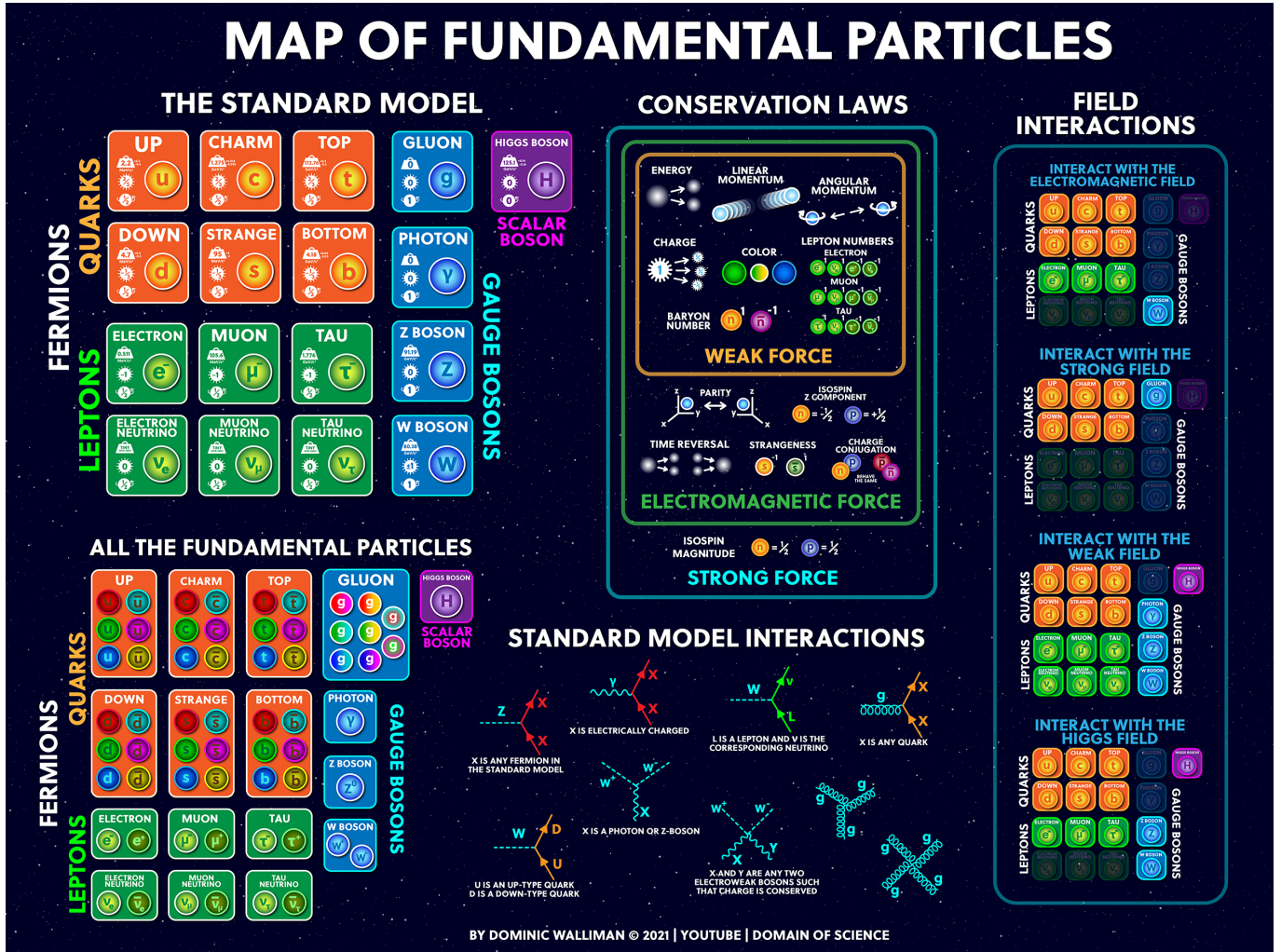


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1 Fundamental Particles



1.1 Units

To simplify calculations we switch to natural units, where $c = \hbar = 1$, $k = 1/4\pi$, and therefore all units are in powers of eV. We can then say that

$$(NU) = \hbar^a c^b (SI)$$

and with that go between systems. Some good conversions to remember is:

$$\hbar c \approx 197 \text{ MeV} \cdot \text{fm}, \quad (1.17)$$

$$\hbar \approx 6.58 \cdot 10^{-16} \text{ eV} \cdot \text{s}. \quad (1.18)$$

2 Scattering and Decay

2.1 Special Relativity

Covariance is a prescription on the form of the physical laws that ensures compliance with the axioms of relativity:

(A1) The physical laws retain the same functional form in all inertial frames.

(A2) The speed of light in vacuum is the same for all inertial observers.

The Four-vector for location is

$$x^\mu = (ct, x, y, z), \quad (2.19)$$

and the transformed vector is:

$$\begin{pmatrix} x'_0 \\ x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma & 0 & 0 \\ \beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad (2.21)$$

$$x'^\mu = \Lambda^\mu_\nu x^\nu. \quad (2.22)$$

The velocity and acceleration four vectors are:

$$v^\mu = \gamma(c, v_x, v_y, v_z),$$

$$a^\mu = \gamma(\dot{\gamma}c, \dot{\gamma}\mathbf{v} + \gamma\mathbf{a}).$$

The four-momentum is:

$$p^\mu = (m\gamma, m\gamma\mathbf{v}) = (E, \mathbf{p}), \quad (2.40)$$

We can use this to calculate energies in particle interactions, since we know there is conservation of momentum.

$$\sum_{i=1}^{N_i} p_i^\mu = \sum_{j=1}^{N_f} p_j^\mu. \quad (2.43)$$

Some helpful equations are:

$$(p^\mu)^2 = (p^0)^2 - |\mathbf{p}|^2, \quad (2.57)$$

$$E = p^0 = (m^2 + |\mathbf{p}|^2)^{1/2}. \quad (2.58)$$

